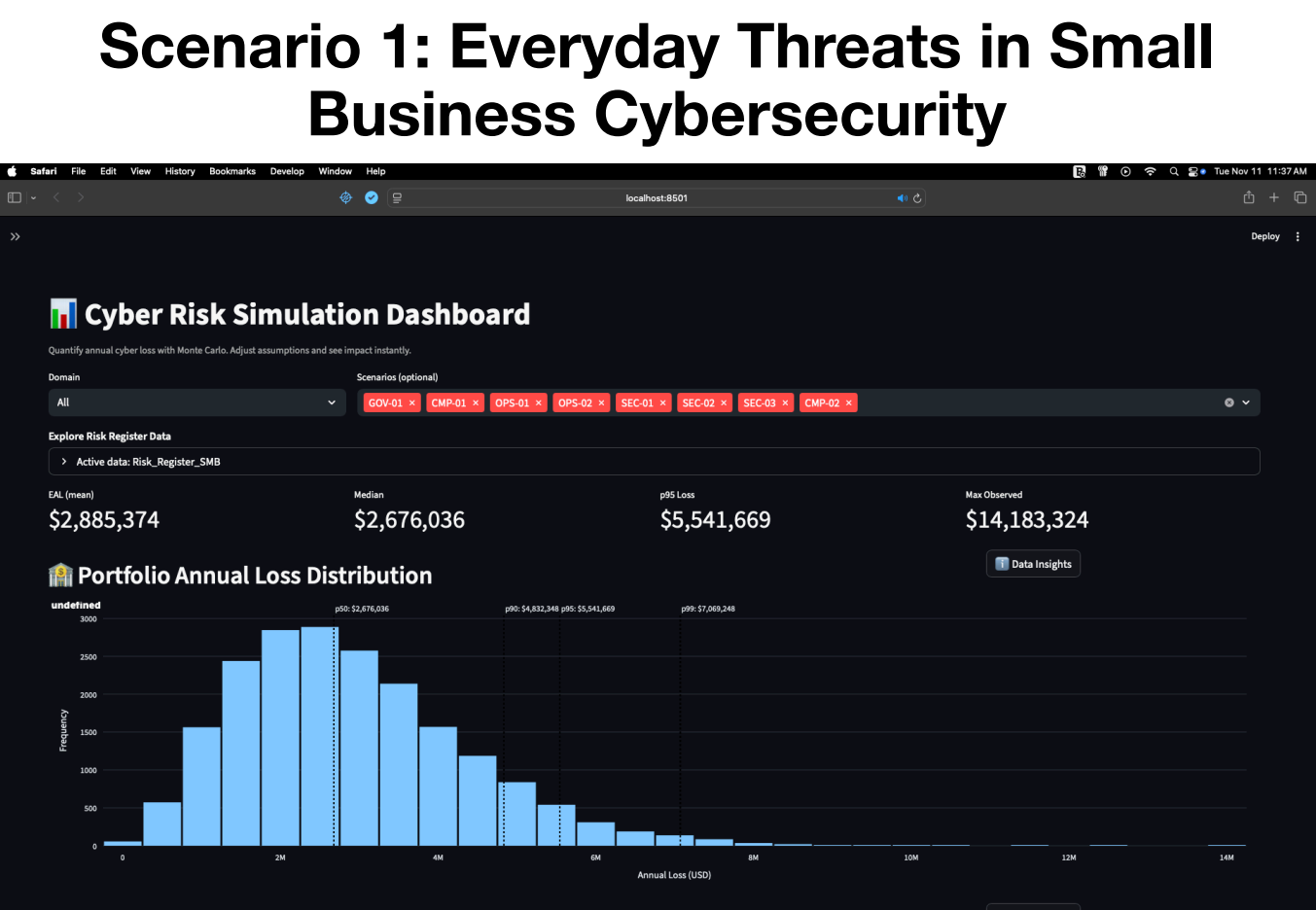


RiskQuant Product Demonstration Series

Annotated Scenario Screenshots and Section References

This slide deck provides annotated screenshots used throughout the RiskQuant scenario demonstrations.

It is designed as a visual companion to the written guides—allowing readers to follow each scenario step-by-step, explore dashboard features, and understand how the quantitative results connect to real-world cybersecurity decisions.



Slides 2-17



RiskQuant Product Demonstration Series

Scenario 1: Everyday Threats in Small Business Cybersecurity

Using financial modeling to understand phishing, stolen credentials, and operational downtime.

2.1 Pre-Run Setup in RiskQuant

1

 Configure Simulation

Number of Simulations

20000 - +

Random Seed

42 - +

> Advanced Settings

2

 Input Data

>  Data Source

3

 Quantify Risk

▶ Run Simulation

↶ Reset

Cyber Risk Simulation Dashboard

Quantify annual cyber loss with Monte Carlo. Adjust assumptions and see impact instantly.

Set parameters in the sidebar and click [Run Simulation](#) to generate results.

2.2 Loading and Exploring the SMB Dataset

Number of Simulations

20000

Random Seed

42

Advanced Settings

Data Source

Select Risk Register

Risk_Register_S...

Reload data

Using risk register:
data/input/Risk_Register_SMB
.xlsx

Cyber Risk Simulation Dashboard

Quantify annual cyber loss Monte Carlo. Adjust assumptions and see impact instantly.

Domain

All

Scenarios (optional)

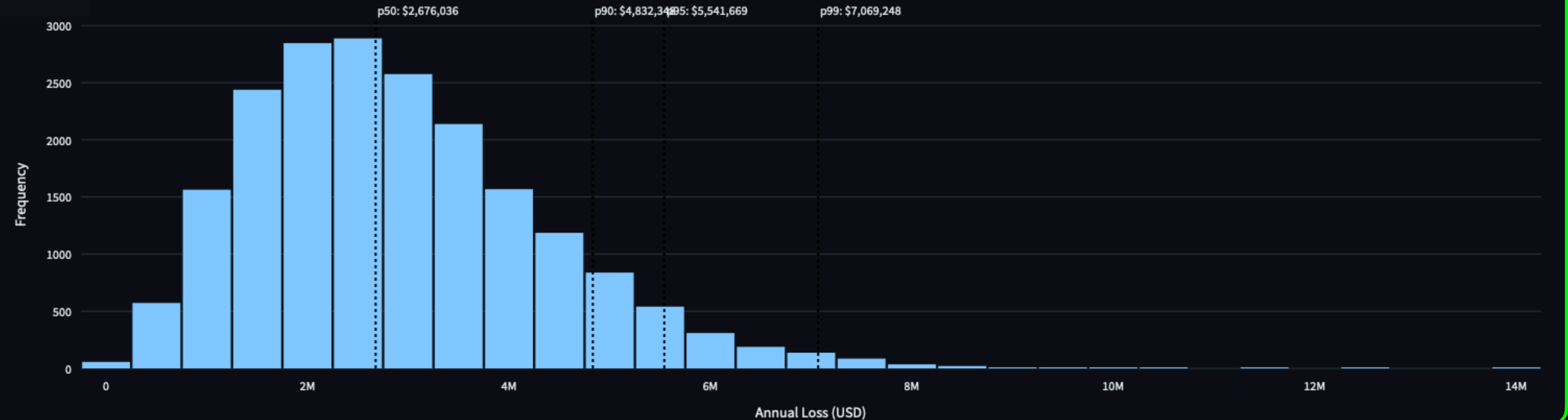
GOV-01 × CMP-01 × OPS-01 × OPS-02 × SEC-01 × SEC-02 × SEC-03 × CMP-02 ×

Explore Risk Register Data

> Active data: Risk_Register_SMB

EAL (mean)	Median	p95 Loss	Max Observed
\$2,885,374	\$2,676,036	\$5,541,669	\$14,183,324

Portfolio Annual Loss Distribution



2.3 Domain Selection



Cyber Risk Simulation Dashboard

Quantify annual cyber loss with Monte Carlo. Adjust assumptions and see impact instantly.

1

Domain

All

All

CMP

GOV

OPS

SEC

Scenarios (optional)

GOV-01

CMP-01

OPS-01

OPS-02

SEC-01

SEC-02

SEC-03

CMP-02

Median

\$2,676,036

p95 Loss

\$5,541,669

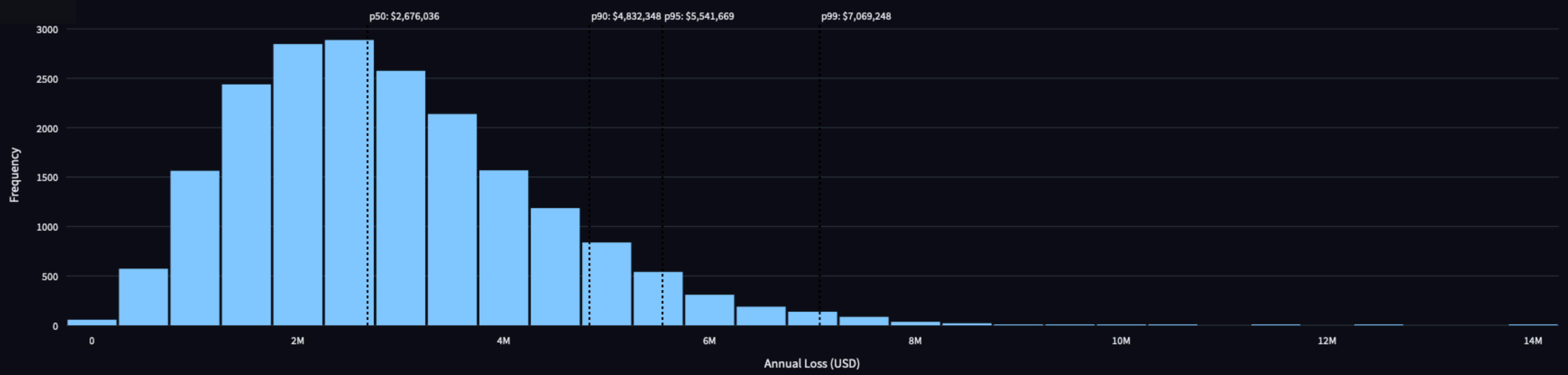
Max Observed

\$14,183,324

Data Insights



Portfolio Annual Loss Distribution



2.4 Filtering and Dynamic Portfolio Refresh

Cyber Risk Simulation Dashboard

Quantify annual cyber loss with Monte Carlo. Adjust assumptions and see impact instantly.

Domain

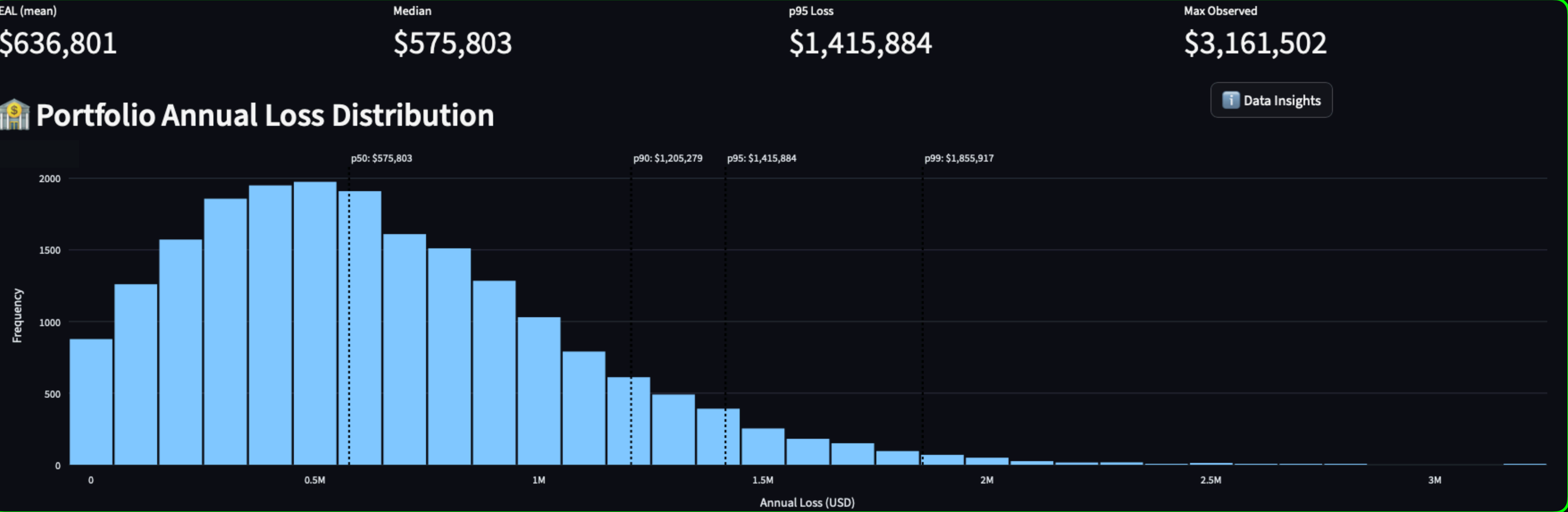
OPS ▼

Scenarios (optional)

OPS-01 × OPS-02 × × ▼

Explore Risk Register Data

> Active data: Risk_Register_SMB



 Portfolio Annual Loss Distribution

i Data Insights



Statistic	Value (USD)
EAL (mean)	\$636,801
Median	\$575,803
p90	\$1,205,279
p95	\$1,415,884
p99	\$1,855,917
Max Observed	\$3,161,502

3.1 Portfolio Annual Loss Distribution - Analyzing the Phishing and Credential Theft Scenario



Deploy

Cyber Risk Simulation Dashboard

Quantify annual cyber loss with Monte Carlo. Adjust assumptions and see impact instantly.

Domain

OPS

Scenarios (optional)

OPS-02

Explore Risk Register Data

> Active data: Risk_Register_SMB

EAL (mean)	Median	p95 Loss	Max Observed
\$289,851	\$261,984	\$682,311	\$1,519,162

Portfolio Annual Loss Distribution

Data Insights



Data Insights Panel—Understanding Portfolio Metrics



Deploy

\$289,851

\$261,984

2 \$682,311

\$1,519,162



Portfolio Annual Loss Distribution



Data Insights

▼ Data Insights — Portfolio Annual Loss Distribution

What it is

Distribution of annualized portfolio loss from Monte Carlo simulations. Each bar shows how often a loss value appears across all runs.

Who it's for

Cyber risk analysts, managers, and decision-makers who need a quick view of typical vs. tail losses to inform risk appetite, budgeting, and control decisions.

How to use it

Compare central tendency (EAL, Median) with tail percentiles (e.g., p95). Use this to set thresholds, evaluate controls or insurance, and communicate exposure.

Definitions

- **EAL (mean):** Expected Annual Loss — arithmetic average of simulated annual losses. **Current: \$289,851**
- **Median:** 50th percentile — half of simulated outcomes are at or below this value. **Current: \$261,984**
- **p95 Loss:** 95th percentile — tail risk indicator; 95% of outcomes are $\leq$ this number. **Current: \$682,311**
- **Max Observed:** Largest annual loss observed in the simulated sample. **Current: \$1,519,162**



Tips

- Adjust simulation runs, percentiles, and scenario filters to see how the distribution and tail change.
- Compare baseline vs. with-control scenarios to demonstrate reduction in tail risk.
- Use the p95 to discuss tail exposure with leadership; use EAL for budgeting.

Close

2500

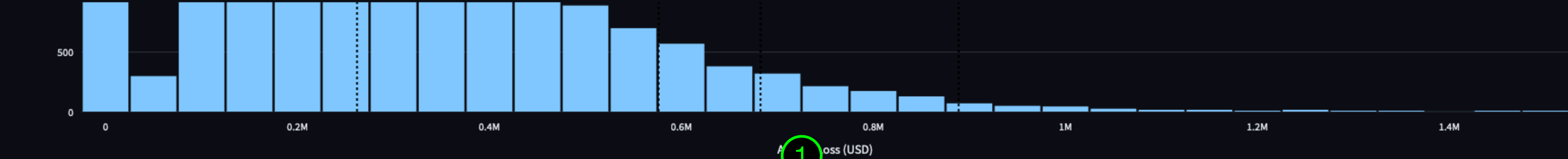
p50: \$261,984

p90: \$576,104

p95: \$682,311

p99: \$888,654

3.3 Cumulative Probability (CDF)—Visualizing Confidence and Risk Thresholds (p95)



Distribution by Scenario

Comparison Type: ☒ Box ☐ Violin

Metric (used when no samples → bar): p95

Sort by: Risk_ID

No row-level samples were returned by the simulator. You can generate per-scenario samples (fast) to enable Box/Violin.

3.3 Cumulative Probability (CDF)—Visualizing Confidence and Risk Thresholds (p50)

>>

Deploy ⋮



Cumulative Probability (CDF)

Data Insights

4

undefined



Distribution by Scenario

Data Insights

Comparison Type

☒ Box ☐ Violin

Metric (used when no samples → bar)

p95

Sort by

Risk_ID

i No row-level samples were returned by the simulator. You can generate per-scenario samples (fast) to enable Box/Violin.

Samples per scenario

3.4 Data Insights Panel – Understanding the Cumulative Distribution Function (CDF)

>>

Deploy



Cumulative Probability (CDF)

Data Insights

▼ Data Insights — Cumulative Probability (CDF)

What it is

The Cumulative Distribution Function (CDF) shows the probability that annual loss will be $\leq$ a given threshold. Right = larger losses; up = higher probability.

Who it’s for

Analysts & leaders setting risk appetite and budget thresholds, and discussing tail risk.

How to use it

Pick a threshold on the x-axis and read up to get $P(\text{Loss} \leq \text{threshold})$. Or pick a probability (e.g., 95%), read left to the curve, then down to the loss at that confidence.

Definitions

- Median (p50): 50% of outcomes are $\leq$ this value. Current: \$261,984
- p90: 90% of outcomes are $\leq$ this value (conservative budgeting). Current: \$576,104
- p95: 95% of outcomes are $\leq$ this value (tail-risk marker). Current: \$682,311
- Displayed percentiles: p50, p90, p95, p99

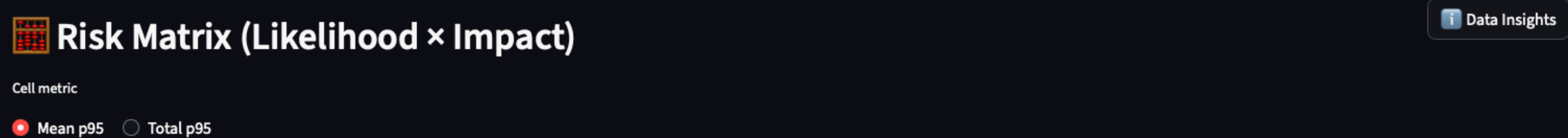


Tips

- Use p95 for tail exposure with leadership; Median/EAL for expected outcomes.
- Steeper curve near your threshold $\Rightarrow$ small control changes can materially change risk.
- Compare baseline vs. with-control CDFs to show tail-risk reduction.

Close

3.5 Distribution by Scenario—Comparing Individual Risks (Box Plot)



>>

Loss Threshold (USD)

 Data Insights

Metric (used when no samples → bar)

p95

Risk_ID 

8

Samples per scenario



 Data Insights

● Mean p95 ○ Total p95

3.6 Data Insights Panel—Distribution by Scenario



0 0.2M 0.4M 0.6M 0.8M 1M 1.2M 1.4M

Loss Threshold (SD)

1



Distribution by Scenario



Data Insights



Data Insights — Distribution by Scenario

What it is

A comparison of simulated annual loss distributions for each scenario. Switch between **Box**, **Violin**, and **Bar** to explore central tendency (P50/median), spread (IQR), and tail risk (outliers/long tails).

Who it's for

- GRC/risk analysts prioritizing scenarios
- Leaders deciding where to invest limited budget

How to use it

1. **Median (P50)** ≈ “typical” outcome; **P95** ≈ “bad but plausible” tail.
2. Wide violins/boxes = more uncertainty; tall bars = higher expected loss.
3. Compare scenarios with similar medians but very different tails—those with higher **P95** may deserve earlier attention.

Definitions

- **P50 (Median)**: 50% of simulations are $\leq$ this value.
- **P95**: 95% of simulations are $\leq$ this value (tail).
- **IQR**: Middle 50% of outcomes (Q1–Q3).
- **Outliers**: Unusually large losses—watch clustering in the upper tail.

Tips

- Sort by **P95** to surface tail-heavy scenarios.
- Use category colors (GOV, CMP, OPS, SEC) to spot themes.
- Re-run with more iterations to stabilize noisy tails.

Comparison Type

☐ Box ☒ Violin

Metric (used when no samples → bar)

p95

Sort by

Risk_ID



No row-level samples were returned by the simulator. You can generate per-scenario samples (fast) to enable Box/Violin.



Generate per-scenario samples (approx.)

Samples per scenario

15000



Export

↓ Download Portfolio Samples (CSV)

 Download Run Manifest (JSON)

[↓ Download Portfolio Samples \(CSV\)](#)

 [Download Run Manifest \(JSON\)](#)

3.8 Data Insights Panel—Risk Matrix (Likelihood x Impact)

1



Risk Matrix (Likelihood × Impact)



Data Insights

▼ Data Insights — Risk Matrix (Likelihood × Impact)

What it is

A qualitative map of **Likelihood × Impact** showing which scenarios are most concerning at a glance. It complements the quantitative charts by offering a quick triage view.

Who it's for

- Executives and managers who need a simple, common-language view
- Analysts aligning portfolio risk to governance processes

How to use it

1. Start triage in the **upper right** (High × High).
2. Cross-check these with the **quantitative tails** (P95) for budget planning.
3. Use labels and contrast toggles to improve readability during reviews.

Definitions & Caveats

- **Likelihood**: Expected frequency (mapped from ARO/λ).
- **Impact**: Consequence of a single loss event (mapped from min/mode/max).
- **Caveat**: The grid is **ordinal**, not monetary—always validate with Monte Carlo results.

Tips

- Keep qualitative→quantitative mappings in sync to avoid drift.
- When two items land in the same cell, use **P95** or **EAL** to break ties.
- Track movement over time (controls, incidents) to show improvement.

Cell metric

☒ Mean p95 ☐ Total p95

☒ Boost contrast

☒ Show labels

4.1 Exporting Results—Documenting and Sharing Insights



Deploy



Risk Matrix (Likelihood × Impact)



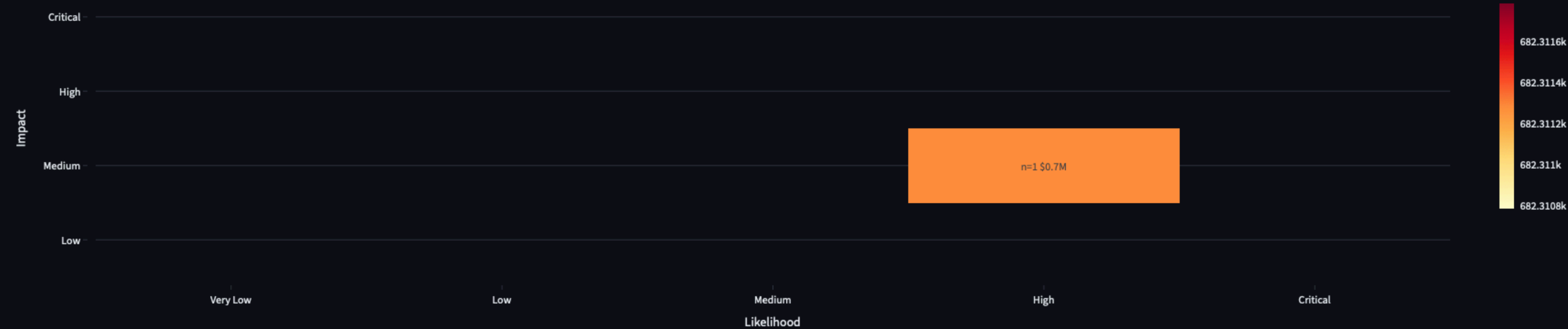
Data Insights

Cell metric

☒ Mean p95 ☐ Total p95

☒ Boost contrast

☒ Show labels



Export

↓ Download Portfolio Samples (CSV)

1



Download Run Manifest (JSON)

2